

Introduction

Motivation:

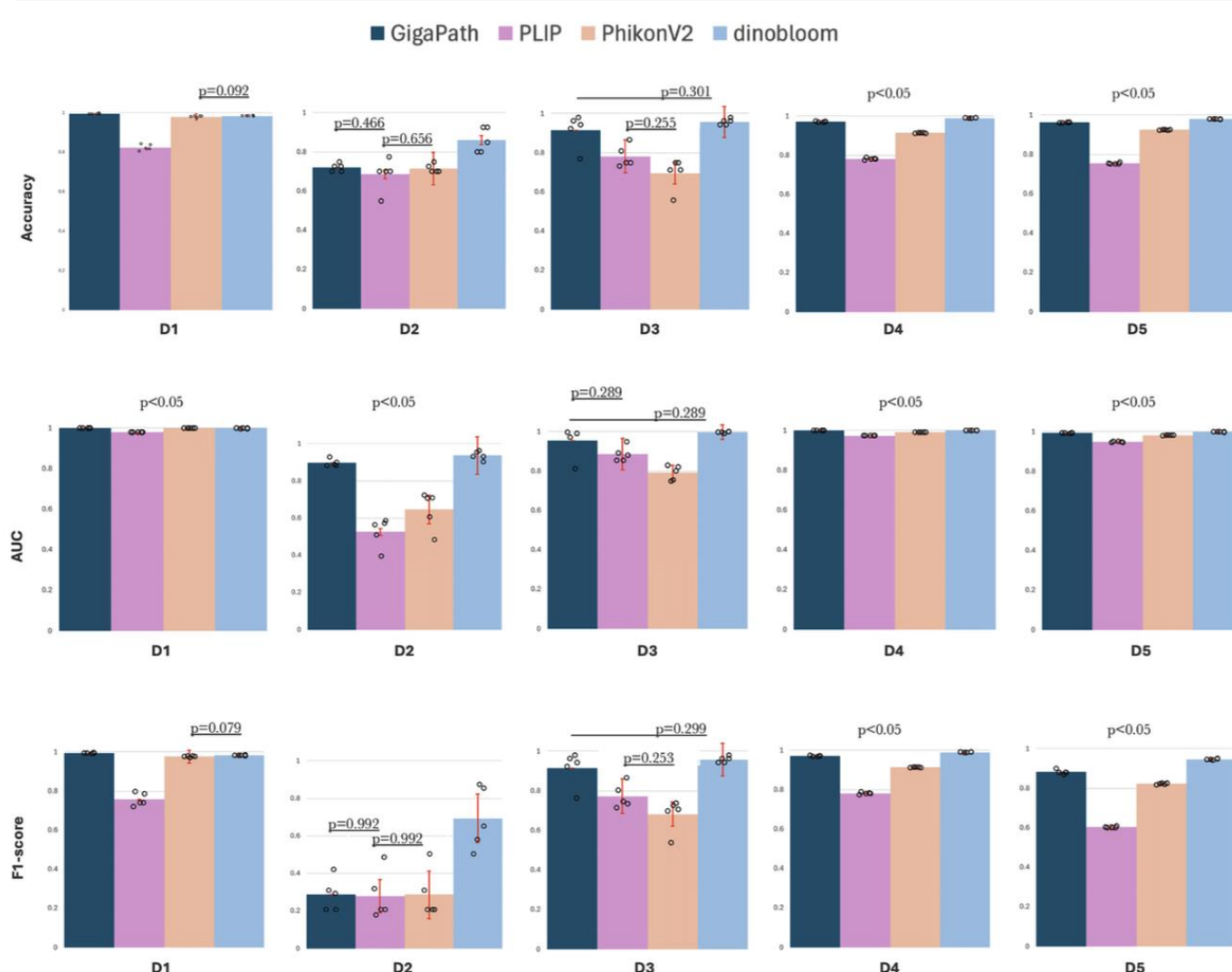
Hematopathology is complex and time-consuming, with a shortage of trained pathologists. AI foundation models (FMs) offer scalable, adaptable solutions, but their domain adaptation to hematology is still underexplored.

Our Contributions:

- **Evaluated** 4 FMs: GigaPath, PLIP, PhikonV2, DinoBloom
- **Benchmarked** on 5 hematology datasets (D1–D5)
- **Compared** 3 learning strategies:
 1. **Feature extraction (S1)**
 2. **Full-data fine-tuning (S2)**
 3. **Partial-data fine-tuning (S3)**
- **Baseline models:** MobileNetV3, ResNet50, ViT

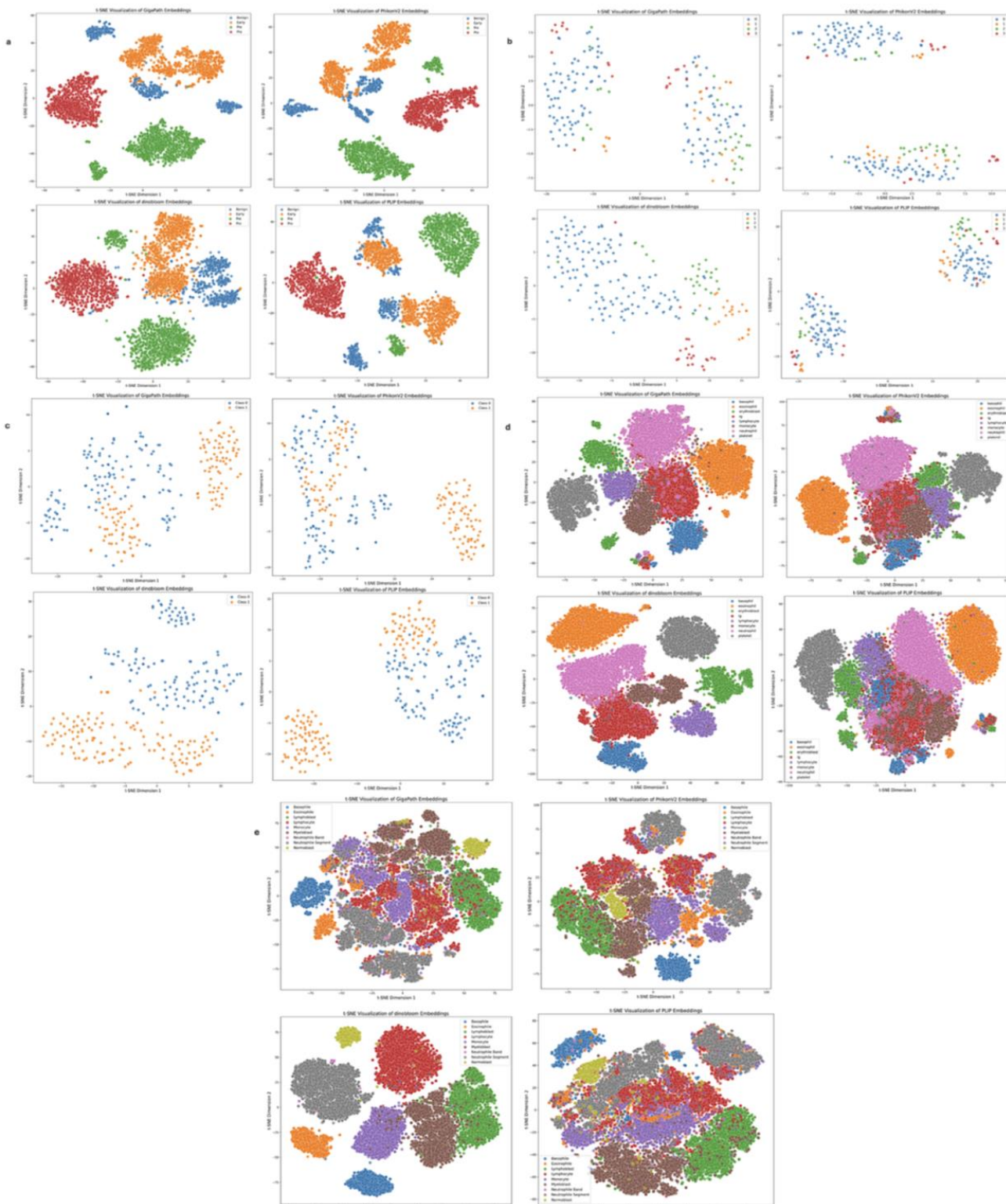
Data & Experimental results

ID	Images	Classes	Class Names
D1 [1]	3,256	4	Benign, Early Pre-B, Pre-B, Pro-B ALL
D2 [2]	300	4	Neutrophil, Lymphocyte, Monocyte, Eosinophil
D3 [3]	260	2	Normal, Malignant
D4 [4]	17,101	8	Neutrophils, Eosinophils, Basophils, Lymphocytes, Monocytes, Immature granulocytes, Erythroblasts, Platelets
D5 [5]	16,037	9	Neutrophil segments, Neutrophil bands, Eosinophils, Basophils, Lymphocytes, Monocytes, Normoblasts, Myeloblasts



- DinoBloom achieved the highest accuracy, AUC, and F1-score on D1, D3, and D4; performance on D1 not significantly different from GigaPath.
- Low results on D2/D3 for all models emphasize dataset size and quality impact.

More experiments and Interpretation



- t-SNE plots show clearer, well-separated clusters from **DinoBloom** across all datasets which highlights its superior ability to produce discriminative feature representations.
- D2 embeddings for all models showed less class separation, confirming its classification difficulty.

Setting	Model	D1			D4		
		Accuracy	F1-score	AUC	Accuracy	F1-score	AUC
Setting 1	GigaPath	0.9942 ± 0.0017	0.9937 ± 0.0019	0.9999 ± 0.0001	0.970 ± 0.002	0.969 ± 0.003	0.999 ± 0.000
	PLIP	0.8227 ± 0.0157	0.7568 ± 0.0331	0.9794 ± 0.0008	0.781 ± 0.005	0.721 ± 0.009	0.975 ± 0.000
	PhikonV2	0.9782 ± 0.0033	0.9767 ± 0.0032	0.9992 ± 0.0001	0.913 ± 0.002	0.900 ± 0.002	0.992 ± 0.000
	Dinobloom	0.9834 ± 0.0023	0.9826 ± 0.0026	0.9986 ± 0.0002	0.989 ± 0.001	0.990 ± 0.001	1.000 ± 0.000
Setting 2	GigaPath	0.982 ± 0.007	0.981 ± 0.007	1.000 ± 0.000	0.9862 ± 0.0046	0.9863 ± 0.0046	0.9998 ± 0.0002
	PLIP	0.960 ± 0.025	0.959 ± 0.025	0.999 ± 0.000	0.9757 ± 0.0077	0.9758 ± 0.0076	0.9995 ± 0.0003
	PhikonV2	0.984 ± 0.005	0.984 ± 0.005	0.999 ± 0.000	0.9867 ± 0.0037	0.9868 ± 0.0037	0.9998 ± 0.0001
	Dinobloom	0.990 ± 0.006	0.990 ± 0.006	1.000 ± 0.000	0.9908 ± 0.0036	0.9908 ± 0.0036	0.9998 ± 0.0001
Setting 3	GigaPath	0.966 ± 0.023	0.965 ± 0.024	1.000 ± 0.001	0.988 ± 0.003	0.988 ± 0.003	1.000 ± 0.000
	PLIP	0.919 ± 0.025	0.916 ± 0.027	0.996 ± 0.002	0.976 ± 0.005	0.976 ± 0.005	0.999 ± 0.000
	PhikonV2	0.947 ± 0.007	0.946 ± 0.008	0.996 ± 0.002	0.982 ± 0.004	0.982 ± 0.004	1.000 ± 0.000
	Dinobloom	0.938 ± 0.025	0.937 ± 0.026	0.989 ± 0.007	0.991 ± 0.001	0.991 ± 0.001	1.000 ± 0.000

- **D1, Setting 1:** GigaPath excelled (Acc: 0.9942, F1: 0.9937, AUC: 0.9999) due to its LongNet architecture for whole-slide processing. DinoBloom closely followed (Acc: 0.9834, F1: 0.9826, AUC: 0.9986).
- **D1, Setting 2:** DinoBloom achieved top results (Acc: 0.99, F1: 0.990, AUC: 1.000).
- **D1, Setting 3:** GigaPath maintained strongest performance (Acc: 0.966, F1: 0.965).
- **D4:** DinoBloom led with Acc/F1 > 0.98 and perfect AUC.

Reference

- 1 Aria, M., Ghaderzadeh, M., Bashash, D., Abolghasemi, H., Asadi, F., Hosseini, A.: Acute lymphoblastic leukemia (ALL) image dataset (2021)
- 2 Zheng, X., Wang, Y., Wang, G., Liu, J.: Fast and robust segmentation of white blood cell images by self-supervised learning. *Micron* 107, 55–71 (2018). <https://doi.org/https://doi.org/10.1016/j.micron.2018.01.010>,
- 3 ALL-IDB Acute Lymphoblastic Leukemia Image Database for Image Processing — scotti.di.unimi.it. <https://scotti.di.unimi.it/all/>, [Accessed 26-05-2025]
- 4 Acevedo, A.: A dataset for microscopic peripheral blood cell images for development of automatic recognition systems (2020)
- 5 Bodzas, A., Kodytek, P., Židek, J.: A high-resolution large-scale dataset of pathological and normal white blood cells (jul 2023)

Conclusion

- DinoBloom achieved the most consistent high performance (Accuracy, F1, AUC > 0.95 in several settings).
- Results vary across datasets and configurations : model choice and tuning are critical for clinical adaptation.
- FMs show strong potential for addressing hematopathology workforce shortages and diagnostic complexity, provided they undergo rigorous validation.